

VOXL 2 Drone Build

Your board is on the bench and talking. It already has flight code, an IMU, six cameras and a live 6S rail — what it does not have is anything that makes thrust. This is the path from here to a first hover.

BENCH STATE — READ FROM THE BOARD

BOARD M0054 ONLINE	SYSTEM IMAGE 1.8.02-14.1a HOLDING	VOXL-SUITE 1.4.0 HOLDING	PX4 Running on SLPI ACTIVE
IMU ICM-42688 / SPI PUBLISHING	POWER RAIL 24.43 V @ 0.09 A 6S LIVE	CAMERAS 5× OV7251 1× IMX214 DETECTED	ESC none on UART MISSING
GPS / MAG never published MISSING	RC LINK ELRS / CRSF link quality 100 LINKED	BOARD NETWORK unknown host OFFLINE	AIRFRAME not built MISSING

STAGE

01

Stay on SDK 1.4 — verify, do not flash

The stack you were shipped already does everything this build needs, and it is calibrated to your specific hardware. Upgrading buys documentation alignment and risks the one thing you cannot regenerate.

An earlier draft of this plan opened with a flash to SDK 1.6.6. Checking the board rather than the changelog overturned that. Three things decide it:

- **Your ESC is already supported.** `vox1-esc 1.4.8` ships `esc_params_generic_m0134_6.xml` — the 6S board in stage 2 is known to the tooling you have. This was the one finding that could have forced an upgrade, and it did not.
- **PX4 1.14.0 is a complete multicopter stack.** Nothing in stages 3–6 needs a newer one.
- **Your calibration is per-unit and irreplaceable.** Stereo front and rear extrinsics and intrinsics, tracking intrinsics, and an IMU calibration all sit in `/data/modalai`. They were measured against your physical

cameras at the factory and cannot be recreated without calibration targets and a rig.

The board also carries 72 offline `.deb` packages in `/data/voxl-suite-offline-packages`, so the missing network route does not block installing anything from the 1.4 set.

Back the calibration up anyway — it costs nothing and it is the only copy:

```
# already done: 13 calibration files + PX4 params -> /home/amit/voxl/backup
adb pull /data/modalai /home/amit/voxl/backup/modalai
adb pull /data/px4 /home/amit/voxl/backup/px4
```

Then confirm the stack is healthy and move straight to stage 2:

```
voxl-version
systemctl is-active voxl-px4
ls /usr/share/modalai/voxl-esc-params/boards/
```

WHEN UPGRADING DOES BECOME THE RIGHT CALL

Three triggers, none of which apply yet: you hit a bug fixed upstream; you post to the ModalAI forum and are asked to reproduce on a current SDK; or you add hardware newer than the 1.4 tooling knows — check `/usr/share/modalai/voxl-esc-params/boards/` before buying. If that day comes, the target is SDK 1.6.6 with system image 1.8.08 (1.6.7 and 1.6.8 are QCS6490-only; 1.7 and 1.8 are alpha and beta). Download `voxl2_SDK_1.6.6.tar.gz` from downloads.modalai.com, run `./install.sh`, answer **preserve** at the `/data` prompt, and do not interrupt it — a killed flash corrupts the filesystem.

STAGE 02

Pick the thrust path, then order

Your ESC choice decides your airframe size, not the other way round.

Your `voxl-px4.conf` is already set to `ESC=VOXL_ESC` and `RC=CRSF_RAW`, so the least-friction build follows that. The constraint is current: ModalAI's 4-in-1 is rated **10A+ per channel**, which is a compact quad, not a 500mm hauler.

Path A — VOXL ESC on UART RECOMMENDED

Use the M0134-6 (2–6S, 5.5–26V, 10A+/channel). Matches your live 24.4V rail and your existing config, and gives you per-motor RPM feedback and current sensing that PX4 actually uses. Build a **250–350mm quad on 5–7" props**. Note the `-6` suffix — the M0134-1 and -3 variants are 4S only and will not survive your 6S pack.

Path B — VOXL 2 IO + third-party PWM ESC

Add the M0065 I/O expander (8× PWM + S.Bus, 25×33mm, 4g) and drive any third-party 6S 4-in-1. Take this only if you want a heavier 450–500mm airframe with 30A+ motors. Costs you the RPM telemetry and needs

`ESC=VOXL2_IO_PWM_ESC` plus a firmware check — boards sold before 2024 need a bootloader update over the debug header.

Parts for Path A, assuming the APM power module already feeding your 24.4V rail stays:

PART	SPEC	NOTE
ESC	M0134-6	4-in-1, 2-6S, UART to J18
Frame	250–350 mm	Carbon X-quad; needs a deck that fits the VOXL 2 footprint
Motors ×4	6S-rated	~1500–1900KV for 5", ~1100–1300KV for 7". Keep continuous draw under the 10A/ch budget
Props	5" or 7"	Match to motor KV; buy spares, you will break them
GPS + mag	u-blox M9N	Holybro or ModalAI unit. Supplies the magnetometer you are currently missing
RC receiver	CRSF	TBS Crossfire Nano RX or ELRS — config is already CRSF_RAW
Battery	6S 1500–3000mAh	Plus a charger and a LiPo-safe bag
Cable	MABL-00081	J18 → ESC. Order ModalAI cables with the boards; they are keyed and fiddly to source later

ORDER THE CABLES WITH THE BOARDS

ModalAI uses JST GH keyed connectors throughout. The single most common way a build stalls for two weeks is having every board on the bench and no cable to join them.

STAGE 03

Wire the airframe

Everything on these headers is 3.3V logic. There is no tolerance for 5V on

a signal pin.

PORT	CARRIES	CONNECT
J4	Main power in, 5V + I2C battery monitoring	APM power module
J18	ESC UART @ 3.3V	M0134-6
J19	GNSS UART + mag I2C + RC UART + 5V	GPS/mag unit and CRSF receiver share this one
J9	USB-C	ADB to your PC — what you are on now
J6 / J7 / J8	Camera groups 0 / 1 / 2, 4-lane MIPI CSI each	Already populated
J2	Fan control, 5V PWM	Cooling fan — fit one, the QRB5165 throttles hot
J10	External SPI expansion	Spare

J19 is the busy one: GNSS, magnetometer and RC all land on that single 12-pin header, so you will be using a breakout pigtail rather than plugging two things in.

VERIFY BEFORE YOU PLUG

Check this map against the [VOXL 2 connector datasheet](#) for your exact board revision before applying power. Confirm motor rotation directions and prop handedness on the bench in stage 5, not in the air.

STAGE 04

Configure PX4

Your config already names the right hardware — this is confirming it against what you actually bolted on.

Re-run the configure pass after the airframe is wired, then check the flight stack settings:

```
vox1-configure-mpa
vox1-configure-cameras

# then edit /etc/modalai/vox1-px4.conf
AIRFRAME=MULTICOPTER
GPS=AUTODETECT
RC=CRSF_RAW          # CRSF_MAV if your RX speaks MAVLink
ESC=VOXL_ESC         # VOXL2_IO_PWM_ESC on Path B
POWER_MANAGER=VOXLPM
```

Confirm the ESC is actually on the bus. This tool stops PX4 and disables its service to take the UART, so restart it afterwards — that is normal, not a fault:

```
vox1-esc detect
vox1-esc calibrate

# restore PX4 when done
systemctl enable vox1-px4 && systemctl restart vox1-px4
```

Then connect QGroundControl over the MAVLink bridge for airframe selection, sensor calibration and RC calibration. `vox1-mavlink-server` is already running on your board.

STAGE 05

Bench bring-up

This is where you find the wiring mistakes, and the only safe time to find them.

PROPS OFF — ALL OF STAGE 5

Remove every propeller before the battery goes in. A 6S quad that arms unexpectedly on a bench will take fingers. Props go on in stage 6 and not before.

Walk the signal chain the same way we walked it today — each of these should now have a live publisher:

```
px4-listener sensor_accel      # already good
px4-listener sensor_mag       # new — arrives with the GPS unit
px4-listener sensor_gps       # needs sky view for a fix
px4-listener input_rc         # timestamp_last_signal must be non-zero
px4-listener battery_status
px4-listener actuator_outputs
```

RC is already done — an ELRS receiver is bound and reporting `link_quality: 100` on J19. The tell is `timestamp_last_signal`, not whether the topic exists. Note that `rsi: -1` and `rc_total_frame_count: 0` are normal on CRSF and are not faults; see `RCSETUP.md`.

Then, still with props off: spin each motor individually and confirm motor order and direction against your airframe's layout, check the mag for interference by watching heading while you throttle up, and verify VIO is tracking with `vox1-inspect-qvio`.

First flight

Short, low, outdoors, and with a plan for putting it down.

- Fresh log enabled, battery failsafe and RC-loss failsafe configured and tested on the bench
- Open space, no people, props inspected and torqued
- First hover in **Stabilized**, not Position — you want to feel the trim before the position controller masks it
- 30 seconds at one metre. Land. Pull the log and read it before the second flight
- Only then move to Position mode, then Offboard for autonomy work

Once it hovers cleanly, the cameras you already have become the interesting part: QVIO for GPS-denied position hold, and the stereo front pair for depth and obstacle avoidance through `voxl-vision-hub`.

Verified against your board over ADB on 15 Sep 2026. Versions and part numbers checked against ModalAI docs the same day.

- [SDK 1.6 release notes](#) — platform support per release
- [Upgrading VOXL SDK](#) — flash procedure
- [VOXL 2 connectors](#) — confirm pinouts here
- [VOXL ESC datasheet](#) — M0134 variants and ratings
- [VOXL 2 IO](#) — Path B expander board
- [PX4 guide: VOXL 2](#)